

BrightLight Tutorials

Data Analytics

Exercise 1: SQL Fundamentals

Database: employees_db

Assume you have a table called **employees** with the following structure:

id	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

Questions

1. SELECT Statement

Write a SQL query to retrieve all columns from the employees table.

2. SELECT DISTINCT Statement

Write a SQL query to find all the unique departments in the employees table.

3. ORDER BY Statement

Write a SQL query to retrieve all employees' first and last names, ordered by salary in descending order.

4. LIMIT Statement

Write a SQL query to retrieve the top 5 highest-paid employees.

5. WHERE Statement

Write a SQL query to find employees who work in the IT department.

6. AND Statement

Write a SQL query to find employees who work in the Finance department **AND** have a salary greater than 58,000.

7. OR Statement

Write a SQL query to find employees who work in the HR department **OR** the Marketing department.

8. NOT Statement

Write a SQL query to find employees who **do not** work in the IT department.

9. IN Statement

Write a SQL query to find employees who are in the HR, IT, or Finance departments.

10. Combining Conditions

Write a SQL query to find employees who are in the IT department, have a salary greater than 50,000, **and** are located in New York.

11. Combining WHERE, AND, and ORDER BY

Write a SQL query to retrieve the first and last names of employees who work in the **Finance** or **Marketing** department, **earn more than 52,000**, and **order the results by salary in descending order**.

12. Combining SELECT DISTINCT, WHERE, and IN

Write a SQL query to find all the **unique cities** where employees work, **excluding those in the IT and HR departments**.

13. Combining WHERE, NOT, AND, and ORDER BY

Write a SQL query to retrieve employees who **are NOT in the Finance department**, **have a salary greater than 50,000**, and **order the results by hire date in ascending order**.

14. Combining WHERE, OR, IN, and LIMIT

Write a SQL query to find the **first 3 employees** who work in either **Chicago** or **Los Angeles** and **belong to the IT or Marketing department**.

SQL FUNDAMENTALS

SELECT STATEMENT

Q1 SELECT *
FROM employees;

ID	First-Name	Last-Name	Department	Salary	hire-date	City
1	John	Doe	IT	55 000	2018-06-15	New York
2	Jane	Smith	HR	48 000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60 000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53 000	2021-03-25	New York
5	David	White	Marketing	52 000	2016-04-10	San Francisco
6	Emily	Davis	IT	62 000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59 000	2019-10-01	Houston
8	Jessica	Moore	HR	51 000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53 000	2022-06-01	Chicago
10	Laura	Hall	IT	50 000	2020-08-10	San Francisco

SELECT DISTINCT STATEMENT

Q2 SELECT DISTINCT department
FROM employees;

department
IT
HR
Finance
Marketing

ORDER BY STATEMENT

Q3 SELECT first-name, last-name, salary
FROM employees
ORDER BY salary DESC;

First-name	last-name	Salary
Emily	Davis	62 000
Mike	Johnson	60 000
Robert	Wilson	59 000
John	Doe	55 000
Sarah	Brown	53 000
Daniel	Clark	53 000
David	White	52 000
Jessica	Moore	51 000
Laura	Hall	50 000
Jane	Smith	48 000

LIMIT STATEMENT

Q4

```
SELECT first_name, last_name, department, salary
FROM employees
ORDER BY salary DESC
LIMIT 5;
```

First-name	last-name	department	Salary
Emily	DAVIS	IT	62 000
Mike	Johnson	finance	60 000
Robert	wilson	finance	59 000
John	Doe	IT	55 000
Sarah	Brown	IT	53 000

WHERE STATEMENT

Q5

```
SELECT *
FROM employees
WHERE department = 'IT';
```

First-name	last-name	department	salary	city
John	Doe	IT	55 000	New York
Sarah	Brown	IT	53 000	New York
Emily	Davis	IT	62 000	Chicago
Laura	Hall	IT	50 000	San Francisco

AND STATEMENT

Q6

```
SELECT *
FROM employees
WHERE department = 'finance'
AND salary > 58 000;
```

First-name	lastname	department	Salary
Mike	Johnson	finance	60 000
Robert	wilson	finance	59 000

OR STATEMENT

Q7

```
SELECT *
FROM employees
WHERE department = 'HR'
OR department = 'marketing';
```

First-name	last-name	department
Jane	Smith	HR
David	White	Marketing
Jessica	Moore	HR
Daniel	Clark	Marketing

NOT STATEMENT

Q8

```
SELECT *
FROM employees
WHERE NOT department = 'IT';
```

First-name	last-name	department
Jane	Smith	HR
Mike	Johnson	Finance
David	White	Marketing
Robert	wilson	finance
Jessica	moore	HR
Daniel	clark	Marketing

IN STATEMENT

Q9
SELECT *
FROM employees
WHERE department IN ('HR', 'IT', 'Finance');

First-name	Last-name	department
John	Doe	IT
Jane	Smith	HR
Mike	Johnson	Finance
Sarah	Brown	IT
Emily	Davis	IT
Robert	Wilson	Finance
Jessica	Moore	HR
Laura	Hall	IT

COMBINING CONDITIONS

Q10
SELECT *
FROM employees
WHERE department = 'IT'
AND salary > 50000
AND city = 'New York';

First-name	Last-name	department	Salary	City
John	Doe	IT	55000	New York
Sarah	Brown	IT	53000	New York

WHERE, AND, & ORDER BY

Q11
SELECT first-name, last-name, department, salary
FROM employees
WHERE department IN ('Finance', 'Marketing')
AND salary > 52000
ORDER BY salary DESC;

First-name	Last-name	department	Salary
Mike	Johnson	Finance	60000
Robert	Wilson	Finance	59000
Daniel	Clark	Marketing	53000

DISTINCT, WHERE & IN

Q12
SELECT DISTINCT city
FROM employees
WHERE department NOT IN ('IT', 'HR');

City
Los Angeles
San Francisco
Houston
Chicago

Q13 COMBINING WHERE, NOT, AND & ORDER BY

SELECT first-name, last-name, department, salary, hire date
 FROM employees
 WHERE department <> 'Finance'
 AND salary > 50000
 ORDER BY hire date ASC;

First-name	last-name	department	salary	hire date
David	White	Marketing	52000	2016-04-11
Jessica	Moore	HR	51000	2016-05-21
John	Doe	IT	55000	2016-06-15
Sarah	Brown	IT	53000	2021-03-21
Daniel	Clark	Marketing	53000	2022-06-01

Q14 COMBINING WHERE, OR, IN & LIMIT

SELECT *
 FROM employees
 WHERE city IN ('Chicago', 'Los Angeles')
 AND department IN ('IT', 'Marketing')
 LIMIT 3

First-name	last-name	department	City
Emily	Davis	IT	Chicago
Daniel	Clark	Marketing	Chicago